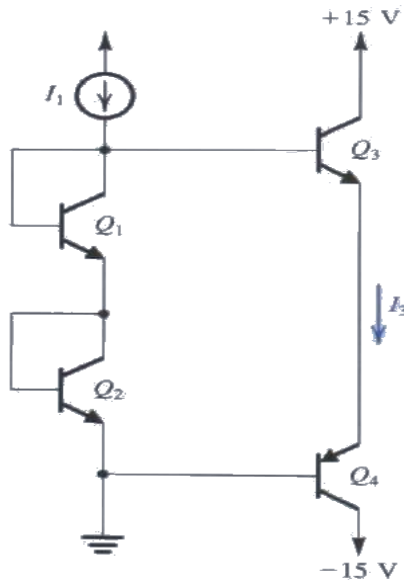


Integrated Circuits

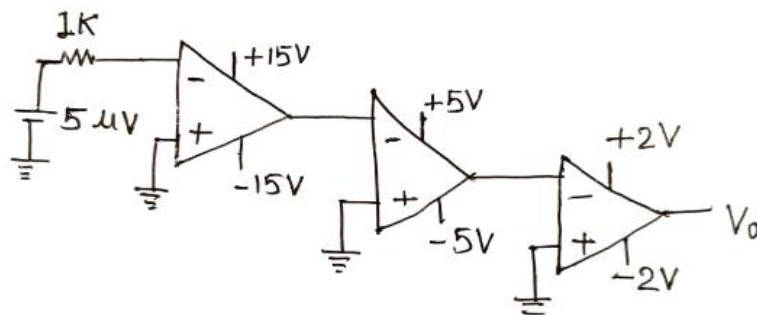
Tutorial/Assignment-01

Q.1: For the circuit shown in the figure below, neglect base currents and use the exponential $I_C - v_{BE}$ relationship to show that

$$I_3 = I_1 \sqrt{\frac{I_{S3} I_{S4}}{I_{S1} I_{S2}}}$$

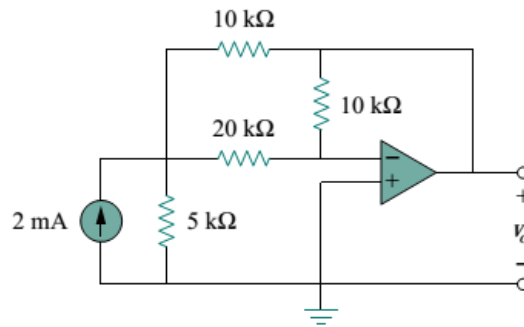


Q.2: Apply the concept of differential gain analysis to obtain output voltage (V_o) in the figure given below



Q.3: An input signal of 0.5 V has to produce undistorted output voltage of 5 V with a fall time of $5\text{ }\mu\text{s}$. Can IC 741 OPAMP be used for such an application. Explain your answer with proper justification [Note : Slew rate of practical Opamp IC 741 is $0.5\text{ V}/\mu\text{s}$]

Q.4: Calculate output Voltage for the circuit shown below



Q.5: If the differential voltage gain and the common mode voltage gain of differential amplifiers are 48 dB and 2 dB respectively, then calculate common mode rejection ratio in dB.

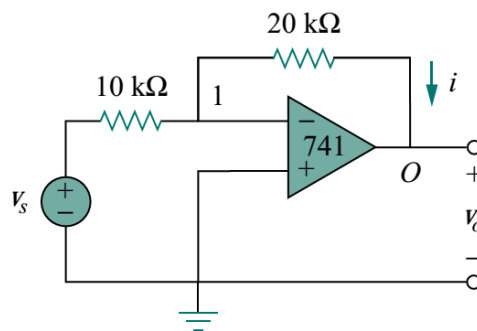
Q.6: Explain DC analysis of input stage of Opamp.

Q.7: Write short note on (i) CMRR (ii) Slew rate (iii) PSRR

Q.8: Explain maximum rating of Opamp.

Q.9: Draw and explain equivalent model of Opamp

Q.10: A 741 Op amp A 741 op amp has an open-loop voltage gain of 2×10^5 , input resistance of $2 \text{ M}\Omega$, and output resistance of 50Ω . The op amp is used in the circuit is shown in fig below.



Determine current I when $V_s = 2\text{V}$